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Crosslinking PDADMAC/PSS polyelectrolyte multilayer membranes for stability at high salinity

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ABSTRACT

Polyelectrolyte multilayer (PEM) nanofiltration (NF) membranes based on PDADMAC (poly (diallyldimethylammoniumchloride)) and PSS (poly (sodium 4-styrenesulfonate)) are known for their high physical and chemical stability. However, under high salinity conditions, the stability of these membranes is compromised due to weakened electrostatic interactions, leading to increased permeability and decreased retention. This study addresses this challenge by crosslinking PDADMAC/PSS multilayers with the photosensitive, negatively charged crosslinker DAS (disodium 4,4′-diazidostilbene-2,2′-disulfonate tetrahydrate). Initially, this crosslinking is studied on model surfaces, demonstrating full stability against desorption by surfactants at high enough DAS concentrations (1 g L^{-1}) and at long enough UV exposure (10 min). Experiments on PEM membranes demonstrate that DAS crosslinking significantly enhanced the stability of PDADMAC/PSS membranes at high salinity, with no permeability increase or loss of selectivity observed up to 1.5 M NaCl, in contrast to non-crosslinked membranes showing a reversible 61 % permeability increase and an irreversible loss in $MgSO_4$ retention of 15 %. At 4 M NaCl, the permeability of non-crosslinked membranes increased by 300 % versus 90 % for crosslinked membranes, again indicating the improved stability of the latter. Crosslinking with DAS further allows tuning of the membrane properties, denser membranes are formed with a lower molecular weight cut-off (MWCO), from around 861 Da of non-crosslinked membranes to around 354 Da of membranes crosslinked with a low DAS concentration (1 g L^{-1}). DAS introduces negative charges (sulfonic acid groups) into the PEMs, changing the membrane charge from positive to highly negative, as evidenced by the high Na_2SO_4 retention (~95 %) and low $CaCl_2$ retention (~7 %) of crosslinked membranes. This study demonstrates the potential of crosslinking with DAS to produce stable PDADMAC/PSS NF membranes with tunable selectivity for challenging separation processes in high-salinity environments.

1. Introduction

In recent years, the Layer-by-Layer (LbL) approach has been demonstrated to be a very versatile method for developing functional nanofiltration (NF) membranes with effective separation properties [1,2]. This technique relies on the self-assembly of polyelectrolyte multilayers (PEMs) onto porous supports by dip-coating the support alternatingly in polycation and polyanion solutions [3]. The multilayer formation process is strongly dependent on the polyelectrolytes' molecular properties (charge density, chain architecture) as well as on the composition of the (aqueous) solvent [4,5]. Hence, this approach enables precise control of surface chemistry and PEM properties by adjusting the coating conditions such as polyelectrolyte (PE) types, the number of layers, the pH and the ionic strength of the coating solution [6,7]. This flexibility allows fine-tuning the performance of these NF membranes for specific/targeted separation processes, going beyond the performance of conventional NF membranes. Such advancements find applications in efficient micropollutant removal [8–10], the development of low-fouling membranes [11,12], the creation of stimuli-responsive membranes [13,14], exploration of organic solvent nanofiltration [15,16], and designing membranes for desalination and selective ion removal [7,17,18].

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In addition to the mentioned advantages, PEM-based NF membranes can exhibit high chemical stability, a crucial factor in broadening the range of applications. In particular, membranes coated with PDADMAC (poly (diallyldimethylammonium chloride)) and PSS (poly (sodium 4-styrenesulfonate)) on negatively charged ultrafiltration (UF) supports demonstrate prolonged physical stability during sequential backwashes and high resistance to hypochlorite cleaning [19]. Furthermore, PDADMAC/PSS-based PEM NF membranes exhibit outstanding, long-term stability under extremely acidic (~ pH 0) and alkaline (~ pH 14) conditions [20]. These findings suggest that these membranes have great potential for application in harsh environments.

In contrast to their stability under acid and alkaline conditions, the performance of PEM NF membranes is compromised when exposed to highly saline feed streams, such as those encountered in water treatment in the petroleum industry, resource recovery from salt lakes and mining industries, and desalination processes [21,22]. Polyelectrolyte multilayers rely on electrostatic interactions that are weakened and even broken at higher salt concentrations. Cheng et al. reported a sharp decrease in $MgCl_2$ retention (from 60 % to 20 %) of the PDADMAC/PSS-based membrane when the feed ionic strength increased from 50 mM to 500 mM NaCl. This reduction was attributed to charge screening by chloride ions and multilayer swelling [23]. Similarly, Han et al. observed significant swelling of PDADMAC/PSS multilayers at high ionic strengths (>1.0 M NaCl) and even a substantial layer dissolution (up to 72 % layer mass loss) of the PEM above 3.0 M NaCl [24]. Additionally, PEMs face challenges when exposed to charged molecules such as surfactants, which can desorb polyelectrolytes from the multilayer through competitive electrostatic binding [25]. Virga et al. demonstrated that around 40 % of the PDADMAC/PSS multilayer can be desorbed by the cationic surfactant CTAB (hexade-cyl-trimethylammonium bromide), while the anionic surfactant SDS (sodium docecyl sulfate) is even capable of removing 60 % of the multilayer, thus compromising the multilayer stability significantly [26]. Hence, high concentrations of salt ions or surfactants, or their combination, can result in compromised membrane performance, leading to increased permeability, reduced separation efficiency, and shorter membrane lifetime.

To improve membrane stability, several studies focused on the chemical crosslinking of PEMs. Glutaraldehyde (GA) has been widely used to crosslink poly (allylamine hydrochloride) (PAH)/PSS multilayers, resulting in denser membranes with enhanced ion selectivities [27,28], membranes with improved stability in the presence of charged surfactants [26] and under acidic conditions [29]. However, GA is typically used to crosslink weak polyelectrolytes containing primary amine groups [30]. Crosslinking the strong polyelectrolyte pair PDADMAC/PSS presents a challenge due to their naturally stable chemical structure (see Fig. 1a and b). An alternative approach involves using the photosensitive crosslinker disodium 4,4′-diazidostilbene-2,2′-disulfonate tetrahydrate (DAS) (Fig. 1c). When exposed to UV light, the azide groups in DAS are converted to highly reactive singlet nitrenes, capable of forming covalent bonds with the polymer matrix through insertion into C–H bonds, addition to double bonds resulting in aziridine rings, or radical abstraction followed by recombination of radical polymer chains [31]. Liu et al. successfully enhanced the monovalent selectivity of anion exchange membranes by crosslinking 2-hydroxypropyltrimethyl ammonium chloride chitosan (HACC)/PSS multilayers with DAS, extending the stable selectivity from 30 h to 72 h [32]. As observed by Restrepo et al., the mechanical stability of biocatalytic polyelectrolyte complex hollow fiber membranes comprising PDADMAC and PSS was enhanced by UV crosslinking with DAS. The crosslinked membrane exhibited a markedly elevated burst pressure, increasing from 0.1 bar to 6 bar [33]. While DAS has shown potential as a crosslinker to enhance the selectivity and mechanical stability of membranes, no studies have investigated crosslinking PDADMAC/PSS multilayers to improve the stability of PEM membranes under high salinity conditions.

In this work, we develop a PEM NF membrane with enhanced stability under high salinity conditions, focusing on the influence of crosslinking on both membrane performance and stability. PDADMAC and PSS, selected for their exceptional chemical stability, are employed as multilayer coatings on a flat sheet poly (ether sulfone) (PES)-based UF support membrane, which is negatively charged through blending with sulfonated poly (ether sulfone) (SPES). This charge improves the adhesion of the first PE layer on the support, further aiding the membrane stability [34]. Crosslinking is achieved using the photosensitive crosslinker DAS and UV irradiation, with a possible reaction mechanism illustrated in Fig. S1. Furthermore, DAS contains two sulfonate groups which provide additional negative charge to the PEMs, allowing adjustment of the membrane charge after layer-by-layer film deposition. Initially, we assess the stability of the PDADMAC/PSS multilayer on model surfaces, before and after crosslinking, by flushing surfactant to the multilayers. Based on the results of Virga et al., who identified SDS as most effective to desorb PEM, this surfactant was used to assess the impact of crosslinking on multilayer stability [26]. Successful crosslinking conditions are then applied to fabricate PEM NF membranes, which are characterized for pure water permeability, molecular weight cut-off (MWCO), and salt retention. The stability of the membranes is addressed by examining the permeability at different ionic strengths (NaCl) before and after crosslinking. This systematic study demonstrates the potential of chemical crosslinking to enhance the stability of PDADMAC/PSS-based multilayer membranes under high salinity conditions and emphasizes the versatility of this fabrication method in producing stable membranes with tunable and efficient separation properties.

2. Experimental

2.1. Materials

Poly (diallyldimethylammoniumchloride) (PDADMAC, M_w = 200,000–350,000 Da, 20 wt%), poly (sodium 4-styrenesulfonate) (PSS,

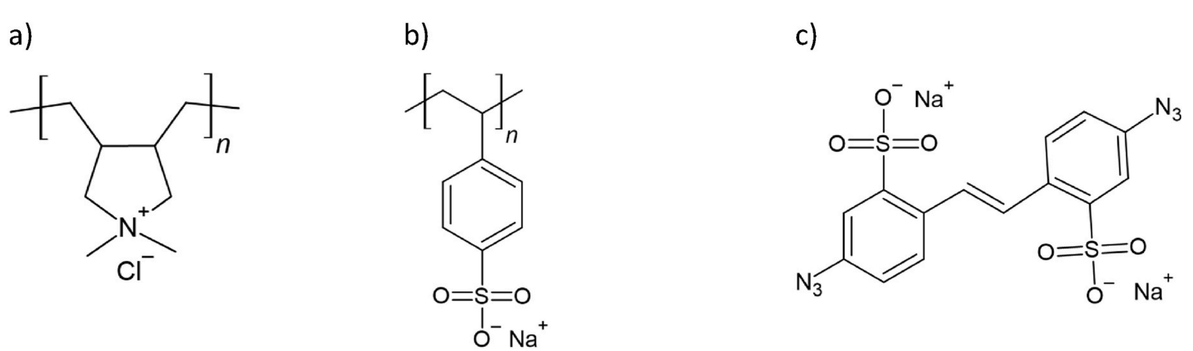

Fig. 1. Chemical structure of a) Poly (diallyldimethylammoniumchloride) (PDADMAC), b) Poly (sodium 4-styrenesulfonate) (PSS), and c) Disodium 4,4′-diazidostilbene-2,2′- disulfonate tetrahydrate (DAS).

$M_w \sim$ 200,000 Da, 30 wt%), ethylene glycol (EG, ReagentPlus, >99 %), triethylene glycol (TEG, ReagentPlus, 99 %), poly (ethylene glycol) (PEG, M_w = 200 Da, 400 Da, 600 Da, 1000 Da, 2000 Da), bovine serum albumin (BSA, M_w = 67,000 Da), PBS buffer (10X Dulbecco's), magnesium sulfate ($MgSO_4$), sodium sulfate (Na_2SO_4), sodium dodecyl sulfate (SDS) and calcium chloride ($CaCl_2$) were obtained from Sigma Aldrich. Disodium 4,4'-diazidostilbene-2,2'- disulfonate tetrahydrate (DAS) (>98 %) was obtained from TCI chemicals. 1-Methyl-2-pyrrolidinone (NMP, 99.5 %) was obtained from Thermal Scientific Chemicals. Glycerol (>99.5 %) was obtained from VWR International. Sodium chloride was obtained from AkzoNobel. Poly (ether sulfone) (PES, Ultrason 6020) and sulfonated poly (ether sulfone) (SPES, GM0559-111-2-T) were kindly provided by BASF. Deionized (DI) water was used throughout this study. All chemicals were used without further purification.

2.2. Polyelectrolyte multilayer stability on SiO_2 model surfaces

2.2.1. Multilayer coating and crosslinking

The silicon wafers underwent an initial cleaning process with piranha solution, which consisted of a 3:1 mixture of concentrated sulfuric acid (H_2SO_4) and hydrogen peroxide (H_2O_2), to remove any organic residues. PDADMAC/PSS multilayers were coated onto the silicon wafers using a dip-coating process, assisted by an in-house fabricated automated coating robot (Fig. S2). First, the wafers were immersed in a 0.5 M NaCl solution for 10 min. The negatively charged silicon wafers were then immersed in 0.1 g L^{-1} PDADMAC solution with an ionic strength of 0.5 M NaCl for 15 min. Subsequently, the wafers were rinsed three times, each for 5 min, in separate baths containing 0.5 M NaCl. These rinsing steps remove weakly adsorbed polyelectrolytes to stabilize the layer and prevent contamination of the subsequent adsorption solution [35]. Following the rinsing steps, the wafers were dipped in a 0.1 g L^{-1} PSS solution with 0.5 M NaCl for 15 min, followed by three further rinses in 0.5 M NaCl. The above steps resulted in the formation of a single bilayer (BL). To achieve the desired number of bilayers, the steps were repeated. All coating steps were executed at room temperature. The nomenclature of the produced PEMs is $(PDADMAC/PSS)_{\#BL}$, with #BL representing the number of bilayers.

For crosslinked PDADMAC/PSS multilayers, an additional step was introduced after the coating process. Specifically, the silicon wafers coated with $(PDADMAC/PSS)_{4.5}$ were immersed in a 0.2 M NaCl, pH 3.8 solution supplemented with DAS (0.1 g L^{-1}, 1 g L^{-1}, 2 g L^{-1}, 5 g L^{-1}, or 10 g L^{-1}) for 4 h to allow DAS molecules to infiltrate the multilayer. Subsequently, the wafers were removed from the DAS solution and exposed to UV irradiation (Gie-Tec, 140007) at 365 nm for a specified duration (1 min, 5 min, 10 min, 30 min, or 60 min). The crosslinked wafers were soaked in a 0.5 M NaCl solution for 24 h to remove excess DAS. Finally, the modified wafers were rinsed with DI water and dried under nitrogen flow. Control silicon wafers coated with $(PDADMAC/PSS)_{4.5}$ were exposed to UV light at 365 nm in the absence of DAS.

2.2.2. Reflectometry

The desorption of multilayers from silicon wafers due to interaction with surfactants was investigated using fixed-angle reflectometry. This technique enables a quantitative and in-situ measurement of the adsorbed and desorbed amounts of multilayer [36]. A solution containing 8 mM of the anionic surfactant SDS and 5 mM NaCl as a background electrolyte was prepared. This SDS concentration exceeds the critical micelle concentration (CMC), as the presence of 5 mM NaCl lowers the CMC of SDS from approximately 8.2 mM–6.8 mM [37]. The SDS solution was flushed onto a silicon wafer pre-coated with $(PDADMAC/PSS)_{4.5}$ under stagnation point flow conditions, ensuring well-controlled hydrodynamics during the multilayer desorption process [36]. After each flushing, the wafer was rinsed with 5 mM of NaCl to desorb the SDS from the PEM and obtain an accurate remaining PEM mass. Both surfactant and rinsing solutions contain 5 mM NaCl as background electrolyte. This process was repeated at least twice until a stable desorption plateau was achieved.

Reflectometry was employed for analysis. A polarised monochromatic light (632.8 nm He–Ne laser) was directed to the wafer surface at the Brewster angle (71°) and reflected toward the detector. The reflected light was split into parallel (p) and perpendicular (s) components, with both intensities (I_p and I_s) continuously and separately measured. The ratio of these two light components, defined as the output signal (S), changes upon adsorption/desorption of polyelectrolyte. Before each measurement, the system was calibrated by flushing with 5 mM NaCl as a solvent to establish a stable baseline (S_0). This change in signal ($\Delta S = S - S_0$) was measured via reflectometry, allowing for the calculation of the adsorbed/desorbed amount according to Equation (1),

$$\Gamma = Q\frac{\Delta S}{S_0} \tag{1}$$

where $\boldsymbol{\Gamma}$ ($mg{\cdot}m^{-2}$) is the amount of mass of PEM adsorbed or desorbed from the silicon wafer, $\boldsymbol{S_0}$ (−) is the starting output signal of the bare wafer. $\boldsymbol{Q}$ ($mg{\cdot}m^{-2}$) is the sensitivity factor which is dependent on the angle of incidence, the thickness of the oxide layer on the silicon wafer, the refractive indices, and the refractive index increment of the adsorbed layer [38]. An optical model was used to determine the $\boldsymbol{Q}$ factor of a multilayer consisting of PDADMAC and PSS on a silicon wafer with a 123 nm silicon oxide layer. The calculated $\boldsymbol{Q}$ factor was 39 mg m^{-2} for the PDADMAC/PSS multilayer. Based on the adsorbed amount of the PE on the silicon wafer, the intrinsic charge density in Coulomb/m^2 of the layer was estimated (see Table S1). All experiments were performed at least in duplicate.

2.2.3. Ellipsometry

Dry and wet PEM thickness was evaluated using a spectroscopic ellipsometer, which measures the change in the polarization state of the propagating light [39]. For this, PDADMAC/PSS multilayers were coated on silicon wafers with a 76 nm SiO_2 top layer in the same way as described in section 2.2.1. To study the effect of crosslinker concentration, $(PDADMAC/PSS)_{4.5}$ were coated on silicon wafers and then crosslinked with 1 g L^{-1}, 5 g L^{-1}, or 10 g L^{-1} DAS at pH = 3.8. First, the dry thickness of the multilayer was measured in situ under 400 mL min^{-1} N_2 flow in a cell until a stable signal was obtained (Mk-2000 X, J. A. Woollam Co., Inc.). Afterwards, the wet thickness of the multilayer was measured in situ using a liquid cell by exposing the wafer to Mili-Q water until a stable signal was obtained (Mk-2000 V, J.A. Woollam Co., Inc.). All measurements were operated at an angle of incidence of 75° and within a wavelength range of 380–1000 nm at room temperature. Optical constants were taken from the software database (Complete Ease). The refractive index was fitted using a standard Cauchy model, according to Equation (2),

$$n(\lambda) = A + \frac{B}{\lambda^2} \tag{2}$$

Where n is the refractive index (−), λ is the wavelength (nm), A and B are Cauchy coefficients. The layer thickness of the PEM was then calculated in the model. In the Cauchy model, the PEM is assumed to be a single layer, and the thickness of the Si and SiO_2 layers underneath the PEM were determined before the actual measurements. In the case of crosslinked layers, the Urbach absorption parameters were fitted to take into account the possibility that the crosslinker DAS can absorb light. For wet measurements, the Cauchy model was extended using the Bruggeman Effective Medium Approximation (BEMA). This approach enabled the differentiation between the two mixed phases, PEM and H_2O. The swelling degree (SD) of the PEM in H_2O was calculated using Equation (3),

$$SD = \frac{d_{wet} - d_{dry}}{d_{dry}} \times 100\% \quad (3)$$

Where d_{dry} (nm) is the dry thickness of the PEM measured under N_2 flow and d_{wet} (nm) is the wet thickness of the PEM measured in DI water. All experiments were performed in triplicate.

2.3. Membrane fabrication

Flat sheet UF membranes were prepared via nonsolvent-induced phase separation (NIPS). The blend polymer solution was prepared by dissolving PES, SPES, and glycerol in NMP at a weight ratio of PES: SPES: glycerol = 14: 7: 12. Glycerol was added to form a thinner skin layer and to achieve a more uniform pore size distribution [40]. The solution was stirred for 48 h at 60 °C to form a homogenous cast solution and then kept in a desiccator for 8 h to degas. A thin film of the obtained polymer solution was cast at room temperature using a casting knife set to a gap height of 150 μm. The film was cast onto a clean glass plate at a controlled speed of 3.5 mm s^{-1}. Immediately following casting, the film was immersed in a coagulation bath containing DI water maintained at 20 °C to induce solidification. Afterwards, the prepared membranes were stored in DI water at 4 °C.

Polyelectrolyte multilayers were coated on top of the prepared UF membranes via the coating process described in section 2.2.1. The resulting PEM membranes were depicted as $(PDADMAC/PSS)_{\#BL}$. For crosslinking, the $(PDADMAC/PSS)_{4.5}$ membranes were immersed in DAS solutions (with varying concentrations of 1 g L^{-1}, 5 g L^{-1}, or 10 g L^{-1} at pH = 3.8) containing 0.2 M NaCl for 4 h. Subsequently, the membranes were taken out from the DAS solutions and exposed to UV at 365 nm for 30 min. The crosslinked membranes were submerged in a 0.5 M NaCl solution for 24 h to remove any excess of DAS. Finally, the modified membranes were stored in DI water at 4 °C before further characterization.

2.4. Membrane characterization

2.4.1. Permeability

The prepared membranes were first characterized for pure water permeability (in $L{\cdot}m^{-2}{\cdot}h^{-1}{\cdot}bar^{-1}$), which was calculated by normalizing the measured pure water flux (J, $L{\cdot}m^{-2}{\cdot}h^{-1}$) for the actual transmembrane pressure, i.e. the applied pressure corrected for osmotic pressure. An Amicon UFSC40001 Stirred Cell with an effective area of 37 cm^2 was used to measure the pure water flux of flat sheet membranes at room temperature and a transmembrane pressure of 2 bar. The membrane's pure water flux J was calculated according to Equation (4),

$$J = \frac{V}{A \Delta t} \quad (4)$$

Where V is the permeate volume (L), A is the effective membrane area (m^2) and Δt (h) is the permeation time.

For the membrane stability tests, the pure water permeability was measured first, followed by the permeability measurement of a NaCl solution with a certain concentration for 2 h, after which the pure water permeability was measured again. These steps were repeated for increasing concentrations of the feed NaCl solution (ranging from 0.5 M to 4 M) with a pure water permeability measurement conducted between each step. All filtration experiments were performed at room temperature and at least in duplicate.

2.4.2. Retentions

The retention performance of the prepared UF membranes was characterized with a 1 g L^{-1} bovine serum albumin (BSA) aqueous solution (in 10 mM phosphate buffered saline (PBS) at pH 7.4). Protein concentrations in permeate and feed were determined spectroscopically at 280 nm using a UV-1800 spectrophotometer (SHIMADZU, Japan), standard solutions were used to construct a calibration curve.

For the non-crosslinked/crosslinked PDADMAC/PSS membranes, single salt retention measurements were performed using the dead-end filtration setup (described in section 2.4.1) with 5 mM solutions of $MgSO_4$, NaCl, Na_2SO_4 and $CaCl_2$. Salt concentrations in the permeate and feed were measured with a conductivity meter (SevenExcellence™, METTLER TOLEDO). All filtration experiments were performed at room temperature in duplicates.

For both BSA and salts, the retention was calculated from the permeate (c_p, mg·L^{-1}) and feed (c_f, mg·L^{-1}) concentrations according to Equation (5).

$$R = \frac{c_f - c_p}{c_f} \times 100\% \quad (5)$$

2.4.3. Molecular weight cut-off

The molecular weight cut-off (MWCO) was measured to estimate the relative changes in pore size of the prepared membranes. For this, a feed mixture of EG, TEG, and PEGs with mean molecular weights of 62, 150, 200, 400, 600, 1000, and 2000 g mol^{-1} was used at a concentration of 1 g L^{-1} each. The permeation experiments were performed in a cross-flow setup with an effective membrane area of 39.5 cm^2 and a cross-flow velocity of 0.258 m s^{-1}, which corresponds to a Reynolds number of around 275. The concentrations in both feed and permeate were analyzed by gel permeation chromatography (GPC, Agilent 1200/1260 Infinity GPC/SEC series) and two size exclusion columns (Polymer Standards Service GmbH, Suprema 8 × 300 mm, 1000 Å, 10 μm followed by 30 Å, 10 μm) in series. Columns were analyzed by a refractive index detector to determine the concentrations. Milli Q water with 50 mg L^{-1} NaN_3 was used as eluent at a flow rate of 1 mL min^{-1}. Compositions were determined through a comparative analysis with a calibration curve established using polyethylene glycol (PEG) standards. The MWCO was defined as the molecular weight at which 90 % retention (10 % sieving) was observed.

2.4.4. Membrane morphology

A scanning electron microscope (SEM) (JEOL JSM-6480LV) was used to capture cross-sectional images of the prepared ultrafiltration membranes. To prevent pore collapse during the drying process, the wet membranes were initially soaked in a solution of 15 wt% glycerol for 4 h. Subsequently, the membranes were taken out from the glycerol solution and dried overnight. Sharp cross-sections were formed by fracturing the membranes directly after immersion in liquid nitrogen. Before imaging, the membrane pieces were sputtered with gold to enhance electrical conductivity (JEOL JFC-1200).

3. Results and discussion

This section is divided into four parts. The first part concerns the effect of crosslinking on the stability of PDADMAC/PSS multilayers on model surfaces against the anionic surfactant SDS, as studied by reflectometry. In the second part, we use identical coating conditions and the obtained crosslinking parameters to develop NF membranes based on PES/SPES support. In part three, we studied the effect of crosslinking on membrane properties regarding MWCO, permeability, swelling, and salt retention. In the final part, we investigated membrane stability when filtration with NaCl solutions of varying ionic strength.

3.1. Effect of crosslinking on the stability of PDADMAC/PSS multilayer against SDS

The growth of PDADMAC/PSS multilayers via layer-by-layer assembly, as well as their desorption upon exposure to the anionic surfactant SDS, has been monitored in situ on model surfaces using reflectometry. Here, we particularly focused on crosslinking the positively charged PEM (terminated with PDADMAC) to facilitate the uptake of the negatively charged crosslinker DAS. Fig. 2 illustrates the desorption of non-crosslinked and crosslinked PDADMAC/PSS-based PEMs after exposure to SDS. Initially, the PEM was rinsed with the background electrolyte solution with identical ionic strength as the SDS solution to establish a baseline signal. Following this, the PEM was flushed with SDS solution and then rinsed again with the background electrolyte solution to ensure the complete removal of any residual adsorbed SDS, as detailed in section 2.2.2. A stable desorption plateau was considered to be achieved after two rounds of alternating flushing and rinsing. The signal difference observed between the flushing and rinsing steps was attributed to the distinct refractive indices of the two solutions. To eliminate the influence of these refractive index variations, the desorbed amount was determined by calculating the mass difference between the initial baseline and the final rinsing plateau, as illustrated in Fig. 2a. It is shown in Fig. 2a that approximately 14 $mg \cdot m^{-2}$ of $(PDADMAC/PSS)_{4.5}$ was desorbed from the silicon wafer, which represents 33 % of the initial total adsorbed amount of $(PDADMAC/PSS)_{4.5}$ (42.3 $mg \cdot m^{-2}$, obtained from Fig. S3). Given that the concentration of SDS in both the bulk solution and at the interface exceeds the CMC of 6.8 mM (section 2.2.2), micelles are formed on the PEM surface. The hydrophilic sulfonic head groups on the micelle surface provided numerous interaction sites for binding with positively charged PDADMAC, leading to partial PEM disassembly [41]. When PDADMAC is desorbed by the SDS micelles, an excess of negative charge would build up in the PEM [26,42]. This excess negative charge would hinder additional adsorption of SDS, thus, the PEM cannot be completely removed from the silicon wafer.

The removal of the PEM is based on weakening the electrostatic interactions between PDADMAC and PSS. We hypothesized that the introduction of covalent bonds between PDADMAC and PSS enhances stability and largely reduces PEM desorption. Fig. 2b demonstrates that following crosslinking with 10 $g \cdot L^{-1}$ DAS, just a negligible quantity of PEM was desorbed. The formation of covalent bonds between PEM layers strengthens the connection of PEs and with that prevents PEM desorption, leading to improved stability.

To better understand the optimal conditions for crosslinking, we investigated PEM stability as a function of crosslinker (DAS) concentration and UV exposure time (Fig. 3). PEM stability here is defined as the percentage ratio of the remaining mass of a PEM and its original mass.

Fig. 3a shows that the lowest concentration of DAS (0.1 $g\ L^{-1}$) did not improve the stability of the PEM after 30 min of UV exposure. Increasing DAS concentration resulted in a significant enhancement of PEM stability. Almost 100 % stability was achieved already at a concentration of 1 $g\ L^{-1}$ DAS. A similar trend was observed with different durations of UV exposure in that stability enhancement requires a minimum exposure time to UV. As shown in Fig. 3b, at the highest DAS concentration of 10 $g\ L^{-1}$, 5 min is still too short to obtain improved stability, whereas nearly 100 % stability was achieved after exposure to UV for 10, 30, and 60 min.

As shown in Fig. 3a, in the absence of crosslinker (0 $g\ L^{-1}$), 30 min exposure to UV resulted in a stability of around 75 %, the same as seen in Fig. 2a–ie for a non-crosslinked PEM that was not exposed to UV at all. Clearly, UV exposure itself does not affect PEM stability. In conclusion, the combination of a sufficiently high concentration of crosslinker (at least 1 $g\ L^{-1}$) and a sufficiently long exposure time to UV light (at least 10 min) significantly enhances the stability of the PEM against SDS.

3.2. Development of PEM NF membranes

In the previous section, we demonstrated that chemical crosslinking can enhance the stability of $(PDADMAC/PSS)_{4.5}$ multilayers. To extend this concept from model surfaces to membranes, we prepared PDADMAC/PSS-based NF membranes based on the obtained crosslinking parameters. Considering the feasibility of employing UV on the membrane surface, the membranes were designed with a flat-sheet geometry. To enhance the overall stability of the membrane, negatively charged SPES was blended into PES to obtain an ultrafiltration support membrane bearing a high ionic surface charge. According to de Grooth et al., the PDADMAC/PSS-based NF membrane with ionic charges (SPES) on the support demonstrated long-term stability against physical backwashes and chemical cleaning with hypochlorite [19]. One possible explanation for this high stability is the improved adhesion of the first layer of positively charged PDADMAC on the support [34]. It adheres more strongly to the highly negatively charged polymeric PES/SPES blend support membrane than it would to an uncharged PES membrane. The highly negative surface charge of the prepared support membrane is reflected by the zeta potential of −35 mV, shown in Fig. S4. The value is comparable with those reported in the work of others [19].

For the selection of a UF membrane suitable for PEM coating, pure water permeability and BSA retention of PES/SPES blend membranes of various polymer concentrations were assesed (see Fig. 4a). The correlation between increasing polymer concentration and decreased pure water permeability suggests a shift towards denser membranes. BSA retention reached near 100 % at the lowest polymer concentration of 21 %, which indicates that the pore size of the membrane was small enough

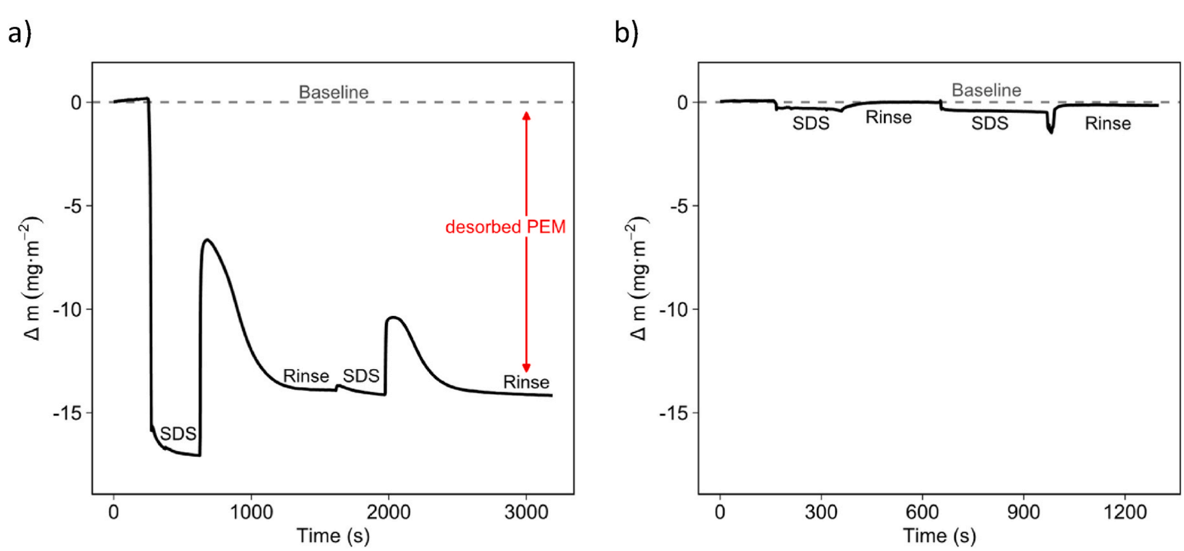

Fig. 2. Desorbed mass of PEM ($mg \cdot m^{-2}$) after flushing with SDS and rinsing solution for a) non-crosslinked $(PDADMAC/PSS)_{4.5}$, and b) $(PDADMAC/PSS)_{4.5}$ crosslinked with 10 $g \cdot L^{-1}$ DAS and UV exposure time of 30 min. Results were obtained via reflectometry.

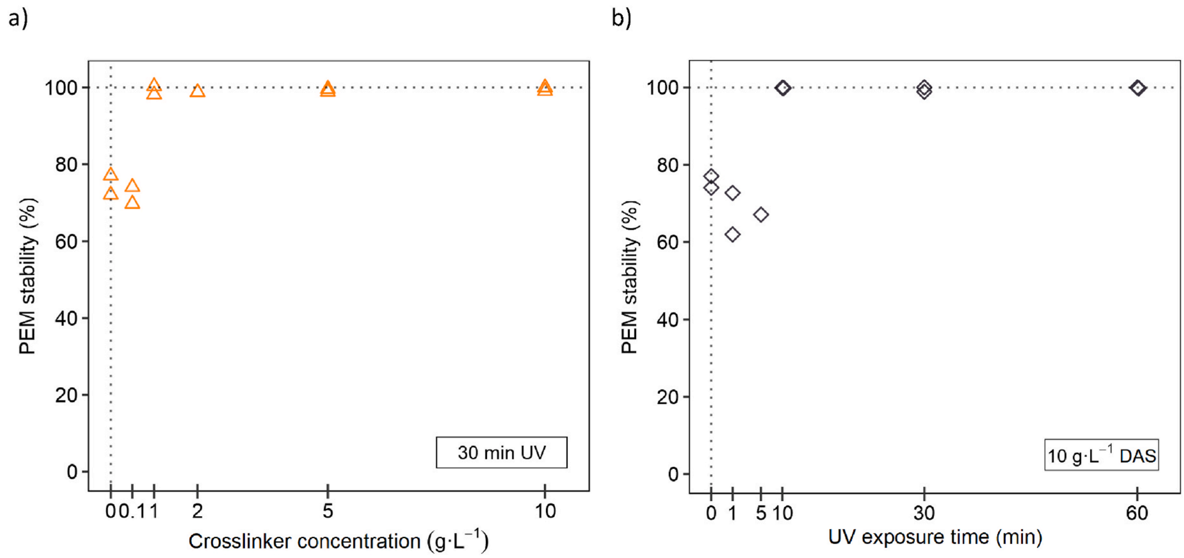

Fig. 3. PEM Stability (%) as a function of: a) Crosslinker (DAS) concentration, at 30 min UV exposure time, or b) UV exposure time, at 10 g L^{-1} crosslinker concentration, of a $(PDADMAC/PSS)_{4.5}$ PEM after flushing with SDS solution and rinsing solution for two rounds. Results were obtained using reflectometry. Data points present individual measurements.

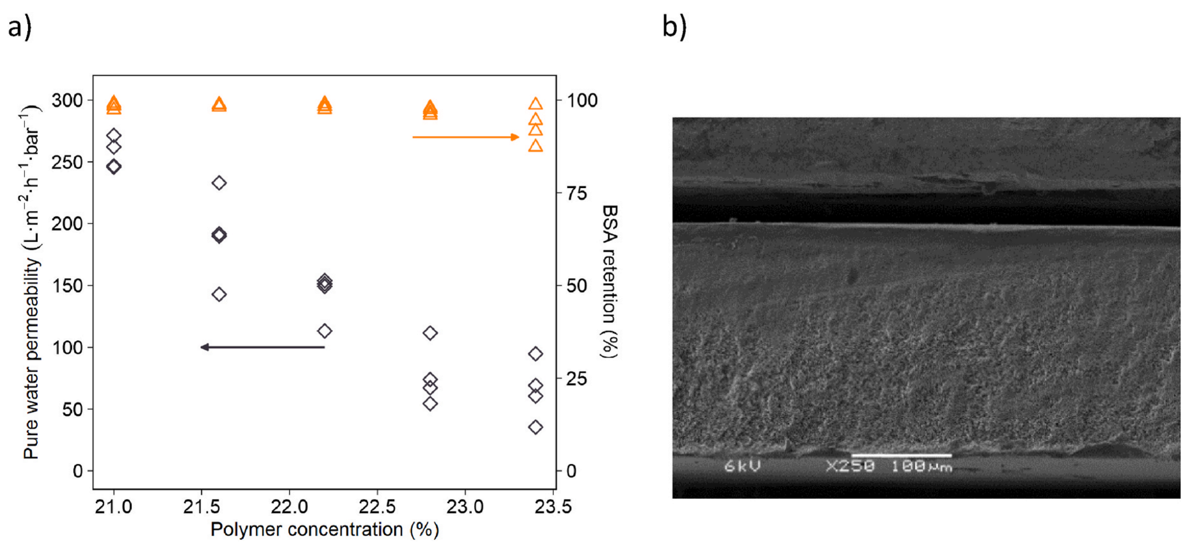

Fig. 4. a) Pure water permeability (L·m^{-2}·h^{-1}·bar^{-1}) and BSA retention (%) as a function of polymer concentration (%) of the PES/SPES blend ultrafiltration membranes with a PES/SPES ratio of 2:1. Data points present individual measurements. b) Cross-section morphology of a PES/SPES blend ultrafiltration membrane with a total polymer concentration of 22.2 %, results obtained by scanning electron microscopy.

to reject almost all the BSA molecules. Therefore, a further increase in the polymer concentration did not lead to an improvement in BSA retention, despite causing the membrane to become denser. At higher polymer concentrations of 22.8 % and 23.4 %, BSA retention became unstable and even slightly decreased, which might be due to the formation of small defects during the phase separation process. These defects likely stem from inhomogeneities in the casted film that we observe at high casting solution viscosities, shown in Fig. S6. The cross-sectional morphology in Fig. 4b demonstrated the absence of macroscopic voids in the membrane at a polymer concentration of 22.2 %, indicating a potential for high membrane mechanical stability. In contrast, the membranes with lower polymer concentrations have large macro-voids in the substructure (Fig. S5). Because of the considerable permeability (141.9 ± 9.6 L m^{-2} h^{-1}·bar^{-1}), high BSA retention (>98 %), and potentially high mechanical stability, the polymer concentration of 22.2 % was selected and from hereon employed as support membranes for subsequent modifications to obtain the PEM NF membranes.

The coating of 4.5 bilayers of PDADMAC/PSS was achieved via layer-by-layer deposition on the prepared support membrane. Results in Table 1 show that the permeability of a $(PDADMAC/PSS)_{4.5}$ membrane (terminated with a positively charged PDADMAC layer) is higher than that of a $(PDADMAC/PSS)_5$ membrane (terminated with a negatively charged PSS layer), 10.4 ± 0.2 L m^{-2} h^{-1}·bar^{-1} and 7.0 ± 0.1 L m^{-2}

Table 1
Pure water permeability and $MgSO_4$ retention of $(PDADMAC/PSS)_{4.5}$ and $(PDADMAC/PSS)_5$ membranes. Values are shown with standard errors based on duplicates.

	Pure water permeability (L·m^{-2}·h^{-1}·bar^{-1})	$MgSO_4$ retention (%)
$(PDADMAC/PSS)_{4.5}$	10.4 ± 0.2	55.5 ± 0.2
$(PDADMAC/PSS)_5$	7.0 ± 0.1	66.2 ± 4.3

$h^{-1} \cdot bar^{-1}$, respectively. In terms of $MgSO_4$ retention, the trend is vice-versa. We attribute this phenomenon to the higher mobility of PDADMAC compared to that of PSS in the PDADMAC/PSS multilayers [43,44]. For PDADMAC-terminated multilayers, an increase in both water and PDADMAC mobility was reported by other works [44–46]. The increase in mobility allows deeper penetration of PDADMAC and water in the PEM and thus leads to excess positive charge, higher swelling degree, and a more open multilayer [43,47]. As a result, the membrane terminated with PDADMAC exhibits higher water permeability and lower $MgSO_4$ retention.

Water transport and salt retention are key characteristics of PEM-based NF membranes. Furthermore, an investigation of the effect of the terminating layer on the rejection of different salt ions adds to a comprehensive understanding of the PDADMAC/PSS-based NF membranes. For this, we measured the retentions $(PDADMAC/PSS)_{4.5}$ and $(PDADMAC/PSS)_5$ membranes for three different ion pairs, namely $CaCl_2$, NaCl, and Na_2SO_4. The results are shown in Fig. 5.

The retention of ions can result from various (combined) mechanisms, of which Donnan exclusion and dielectric exclusion are most prominent in charged NF membranes [48]. Donnan exclusion relies on the balance between the repulsion of co-ions and attraction of counter-ions at the charged membrane interface while maintaining electroneutrality in the permeate, while dielectric exclusion is the process where differences between membrane and bulk dielectric constants cause ions to be repelled due to interactions with bound charges induced by dielectric polarization [49]. For the negatively charged PSS-terminated $(PDADMAC/PSS)_5$ membrane, the 88 % retention of Na_2SO_4 (with a bivalent co-ion SO_4^{2-} and monovalent counter-ion Na^+) was higher than the 27 % retention of $CaCl_2$ (with a divalent counter-ion Ca^{2+} and monovalent co-ion Cl^-). The opposite trend was observed for the positively charged PDADMAC terminated $(PDADMAC/PSS)_{4.5}$ membrane, with a higher retention of $CaCl_2$ (56 %) than Na_2SO_4 (38 %). This observed retention trend for both membranes aligns well with the Donnan exclusion model, indicating a highly negatively charged $(PDADMAC/PSS)_5$ membrane and a slightly positively charged $(PDADMAC/PSS)_{4.5}$ membrane [48]. For both membranes, NaCl exhibited the lowest retention (<20 %). The higher retention of $CaCl_2$ compared to that of NaCl for the negatively charged $(PDADMAC/PSS)_5$ membrane may be attributed to the smaller hydrated ion size and a higher diffusion coefficient of the monovalent ion Na^+ than the divalent Ca^{2+} (see Table 2) [48].

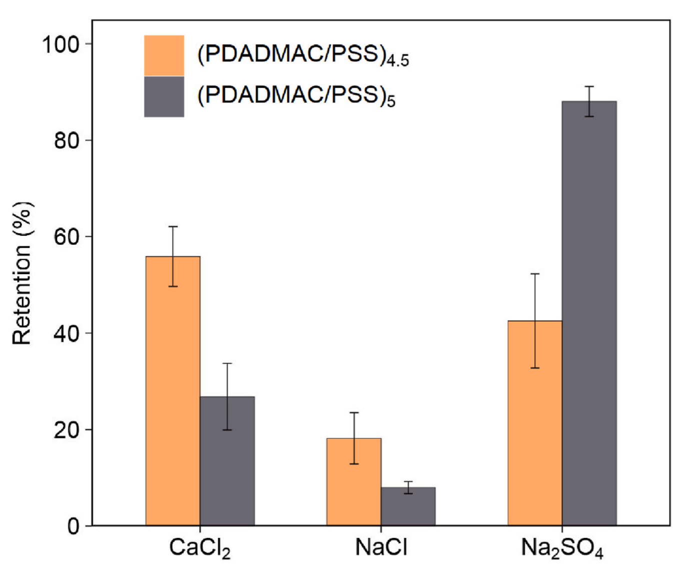

Fig. 5. Retentions of 5 mM $CaCl_2$, NaCl, and Na_2SO_4 of $(PDADMAC/PSS)_{4.5}$ and $(PDADMAC/PSS)_5$ membranes. Error bars display the standard error (sample size n = 2).

Table 2
Bare- and hydrated ion sizes (radius) and ionic diffusion coefficients [50,51].

	Bare ion radius (A^0)	Hydrated ion radius (A^0)	Bulk diffusion coefficient ($x10^{-9}$ m^2/s)
Na^+	0.95	3.58	1.33
Ca^{2+}	1.00	4.12	0.92
Cl^-	1.81	3.32	2.03
SO_4^{2-}	2.90	3.79	1.06

3.3. Effect of crosslinking on membrane properties

So far our findings indicate that, on model surfaces, the stability of the PDADMAC/PSS-multilayer can be improved by UV-induced chemical crosslinking with DAS. Next, we prepared flat-sheet PEM NF membranes based on PDADMAC/PSS to extend the insights from the study on model surfaces to the chemical crosslinking of the positively charged PDADMAC-terminated membrane. For this, we prepared and studied non-crosslinked $(PDADMAC/PSS)_{4.5}$ membranes as well as $(PDADMAC/PSS)_{4.5}$ membranes crosslinked with either 1 g L^{-1}, 5 g L^{-1} or 10 g L^{-1} DAS.

The retention of uncharged organic molecules, such as polyethylene glycols (PEGs) with a molecular mass higher than the molecular weight cut-off (MWCO) of membranes, is primarily driven by physical sieving through the pores [52]. Therefore, we measured the MWCO of the membranes by filtering different-sized PEGs to describe the effective pore size of the membranes. As shown in Fig. S7, some of the membranes show leakage of large-sized PEGs, as indicated by sieving curves that did not approach zero near 2000 Da. Assuming that defects are responsible for this behavior, in order to compare the pore sizes of just the defect-free membrane area, we normalized the sieving coefficients of the membranes at 2000 Da to zero and adjusted the sieving curves accordingly to calculate 90 % MWCO. The determined MWCO and pure water permeability are shown in Fig. 6a. Crosslinking with 1 g L^{-1} DAS resulted in a significantly lower MWCO of 354 ± 20 Da compared to 861 ± 22.5 Da of the non-crosslinked membranes. This suggests that crosslinking led to a denser structure and smaller mesh/pore sizes in the polymer matrix. A clear difference in pure water permeability was observed between the more open non-crosslinked membranes and the denser crosslinked membranes. The pure water permeability of the non-crosslinked membranes was found to be 10.4 ± 0.2 L m^{-2} $h^{-1} \cdot bar^{-1}$, whereas for the 1 g L^{-1} DAS crosslinked membranes, it decreased to 7.4 ± 1.2 L m^{-2} $h^{-1} \cdot bar^{-1}$ due to a smaller pore size. The MWCO increased with increasing DAS concentration. At a concentration of 5 g L^{-1} DAS, the MWCO was 429 ± 66 Da, while at a concentration of 10 g L^{-1} DAS, the MWCO further increased to 490 ± 10.5 Da. This, somewhat counter-intuitive observation, can be attributed to the increased swelling degree of the PDADMAC/PSS multilayer, as shown in Fig. 6b. The swelling degree of the PEM in H_2O was determined based on the dry/wet thicknesses measured on model surfaces via ellipsometry (see Table S2). With increasing DAS concentration, the swelling degree increased from around 80 %–95 %, and further to approximately 108 % [44].

To understand the changes in MWCO, pure water permeability, and swelling degree after crosslinking, it is crucial to recognize that the addition of DAS to the multilayer system induces two simultaneous, yet opposing, effects: a) an increase in crosslinking density, resulting in an denser membrane, and b) the introduction of additional charge due to the sulfonate groups of DAS (see chemical structure of DAS in Fig. 1), which can lead to an increase in swelling in water. The higher the DAS concentration, the greater the negative charge is introduced into the PEM. More excessive charge leads to greater repulsion between free ionic sites and thus a higher swelling degree [43]. At low DAS concentrations (1 g L^{-1}), the crosslinking effect dominates, resulting in denser membranes with lower MWCO and pure water permeability. Conversely, with increasing DAS concentration, the swelling effect

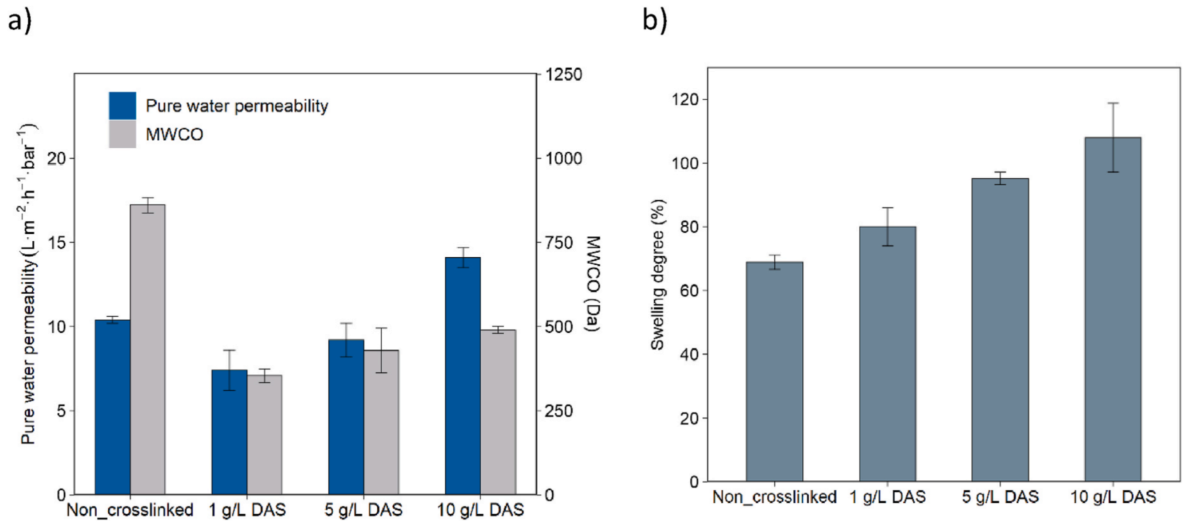

Fig. 6. a) Molecular weight cut-off (MWCO), b) PEM swelling degree in pure H_2O (%), of $(PDADMAC/PSS)_{4.5}$-coated membranes without crosslinking as well as after crosslinking with DAS concentrations of 1 g L^{-1}, 5 g L^{-1} or 10 g L^{-1}. Ellipsometry was used to determine the dry and wet PEM thicknesses and then the swelling degree was calculated accordingly. Error bars display the standard error (sample size n = 2 or 3).

caused by charge becomes more pronounced, leading to a slight increase in MWCO and pure water permeability. The 10 g L^{-1} DAS crosslinked membrane achieves higher pure water permeability compared to the non-crosslinked membrane, while maintaining a much lower MWCO of around 500 Da. This demonstrates that crosslinking with DAS enables both enhanced water transport and a denser polymer network.

Fig. 7 shows the retentions of the non-crosslinked and crosslinked $(PDADMAC/PSS)_{4.5}$ membranes for different ion pairs. As discussed in section 3.2, the positively charged non-crosslinked $(PDADMAC/PSS)_{4.5}$ membrane showed a salt retention sequence of $CaCl_2 > Na_2SO_4 > NaCl$. Upon crosslinking and regardless of the DAS concentration, the salt retention sequence changed to $Na_2SO_4 > NaCl > CaCl_2$, which aligns well with the Donnan exclusion model for negatively charged NF membranes [49]. As shown in Fig. 5, the negatively charged but non-crosslinked $(PDADMAC/PSS)_5$ membrane showed higher retention for $CaCl_2$ than for NaCl. This can be linked to the smaller hydrated ion size and higher diffusion coefficient of Na^+ (see Table 2), suggesting that Donnan exclusion alone does not fully govern ion retention in this membrane, as discussed in section 3.2. Crosslinking the initially positively charged $(PDADMAC/PSS)_{4.5}$ membrane with DAS introduces enough negative charge to achieve a shift in membrane surface charge from positive to negative and turn Donnan exclusion into the dominating retention mechanism. As a result, the crosslinked membranes show higher retention of the monovalent Na^+ than the divalent Ca^{2+}. The lower valence of the counter-ion Na^+ leads to a lower co-ion (Cl^-) concentration in the membrane, which leads to higher retention of NaCl, fully in line with Donan exclusion principles [48].

In conclusion, crosslinking with DAS results in denser membranes, as evidenced by the reduced MWCO due to increased network density. The negatively charged sulfonate groups in DAS play a crucial role in tuning membrane properties, leading to a shift in surface charge from positive to negative and enhanced water transport. Overall, crosslinking with DAS allows for control over membrane properties, balancing selectivity, permeability, and structural stability for tailored performance.

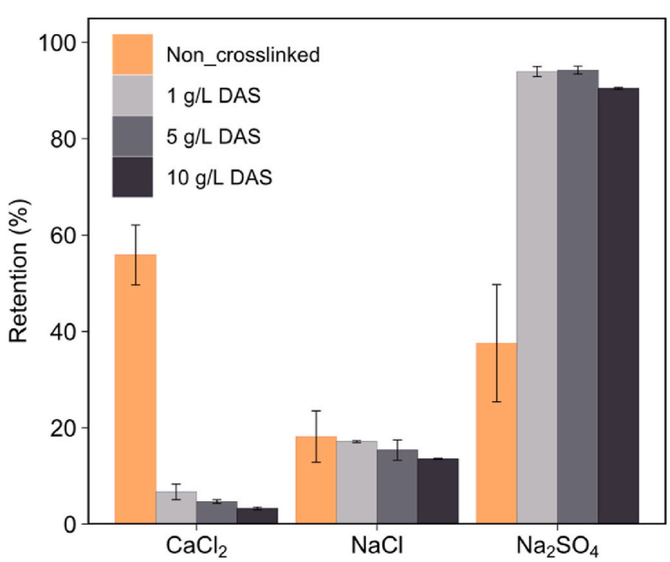

Fig. 7. Retentions of 5 mM $CaCl_2$, NaCl, and Na_2SO_4, of $(PDADMAC/PSS)_{4.5}$-coated membranes without crosslinking as well as after crosslinking with DAS concentrations of 1 g L^{-1}, 5 g L^{-1} or 10 g L^{-1}. Error bars display the standard error (sample size n = 2).

3.4. *Stable crosslinked membranes at high salinity*

In the previous section, we have shown the separation properties of the crosslinked membranes. Here, we investigate the stability of these membranes when subjected to high salinity. To this end, four types of membranes were examined: non-crosslinked $(PDADMAC/PSS)_{4.5}$ membranes, as well as $(PDADMAC/PSS)_{4.5}$ membranes, crosslinked with either 1 g L^{-1}, 5 g L^{-1} or 10 g L^{-1} DAS. The membranes were alternatingly exposed to pure water and NaCl feed solutions with ionic strengths of 0.5 M, 1.0 M, and 1.5 M. The membrane permeability was evaluated during the filtration of each feed solution, as illustrated in Fig. 8a.

Both osmotic pressure and swelling play an important role in the permeability of membranes in salt solutions. The observed permeability decrease of all membranes at 0.5 M and 1.0 M NaCl can be attributed to a result of osmotic pressure-driven dehydration (shrinkage) of the multilayer where salt ions draw out water as the bulk osmotic pressure exceeds that inside the multilayer [22,53]. With increased salt concentration, the non-crosslinked membrane showed a 25 % increase in permeability at 1.0 M NaCl in comparison to the initial water permeability, while an 80 % increase was observed at 1.5 M NaCl. The increase in membrane permeability at high salinity can be attributed to the enhanced mobility and swelling of the PEM, an effect that can be understood in terms of thermodynamics by evaluating the Gibbs free

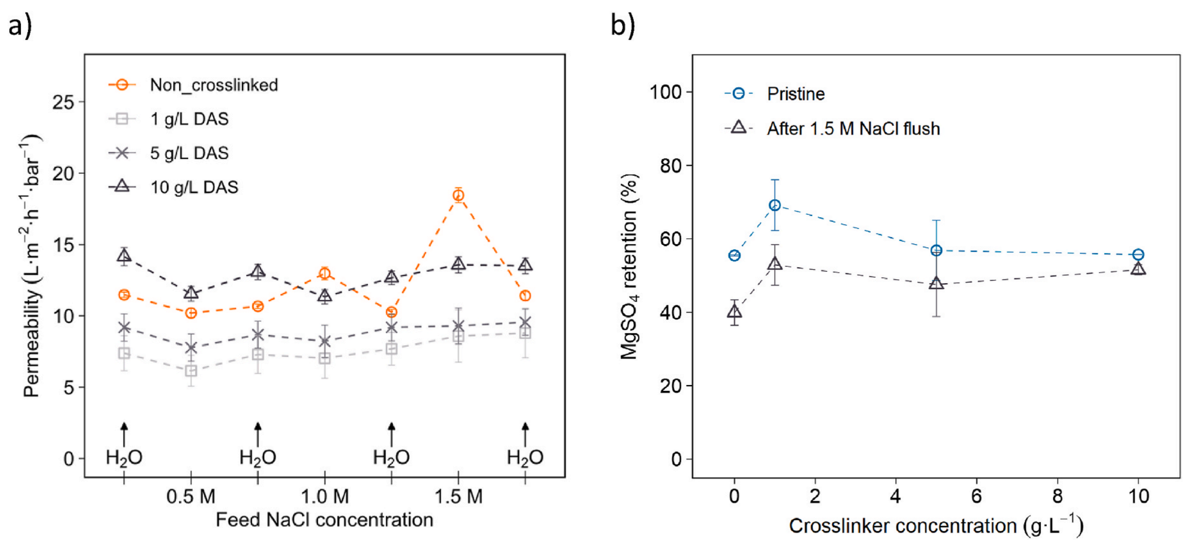

Fig. 8. a) Permeability (L·m^{-2} ·h^{-1}·bar^{-1}) of non-crosslinked and crosslinked $(PDADMAC/PSS)_{4.5}$ membranes during filtration alternately with pure water and 0.5, 1.0, or 1.5 M NaCl solutions, at a DAS concentration of either 1 g L^{-1}, 5 g L^{-1} or 10 g L^{-1}. b) $MgSO_4$ retention of $(PDADMAC/PSS)_{4.5}$-coated membranes before and after exposure to 1.5 M NaCl, as a function of crosslinker (DAS) concentration. Error bars display the standard error (sample size n = 2).

energy change $\Delta_f G$ of PE complexation [54]. The change in enthalpy ($\Delta_f H$) of polyelectrolyte complex formation increases from negative to positive with increasing salt concentration. As for the change in entropy ($\Delta_f S$), this remains always positive but decreases at higher salt concentrations [55]. Given the relationship $\Delta_f G = \Delta_f H - T\Delta_f S$, it can be postulated that at a high enough salt concentration, the enthalpy change exceeds the product of temperature and entropy change ($\Delta_f H > T\Delta_f S$), causing the Gibbs free energy becomes positive ($\Delta_f G > 0$). Consequently, this positive Gibbs free energy will hamper PE complexation. As a result, the mobility of PEMs may increase and eventually, layer rearrangement can occur. This, coupled with other factors such as the increase in osmotic pressure with increasing salt concentration, facilitates the penetration of more external salt ions and water molecules into the multilayer, inducing significant swelling. This ultimately results in an increase in membrane permeability at high ionic strength.

The pure water permeability of the non-crosslinked membrane returned to its initial value after exposure to 1.0 and 1.5 M NaCl (see Fig. 8a). This suggests that the salt-induced PEM swelling is reversible. However, evaluating permeability alone does not justify this conclusion, it becomes evident after analyzing the retention of $MgSO_4$. After exposure to 1.5 M NaCl, the $MgSO_4$ retention of the non-crosslinked (0 g L^{-1} crosslinker) membranes reduced from 55 % to 40 %, when compared to that before filtration with 1.5 M NaCl (see Fig. 8b). This indicates that while permeability changes are reversible, the change in retention is not.

In contrast, crosslinked membranes demonstrated enhanced stability regarding both permeability and $MgSO_4$ retention. Upon increasing the NaCl concentration to 1.0 M, the permeability of the crosslinked membranes remained stable, irrespective of the concentration of DAS. Even at 1.5 M NaCl, membranes crosslinked with 5 g L^{-1} and 10 g L^{-1} DAS showed no increase in permeability. As expected, a cross-linked network significantly reduced layer swelling when exposed to high concentrations of salt ions. This indicates that successful crosslinking of the PEM occurred, enhancing membrane stability in high salinity conditions. In terms of $MgSO_4$ retention, at low DAS concentration (1 g L^{-1}), $MgSO_4$ retention dropped from 70 % (before exposure to 1.5 M NaCl) to 53 % (after exposure to 1.5 M NaCl), as shown in Fig. 8b. However, this drop in $MgSO_4$ retention narrows with increasing DAS concentration, and almost no retention drop was observed for the 10 g L^{-1} DAS crosslinked membrane. This suggests that an increase in DAS concentration increases the amount of formed crosslinks. The growth of crosslinked networks largely inhibits layer swelling in NaCl even at high ionic strengths. Therefore, crosslinking with DAS enhanced the stability of PDADMAC/PSS-based NF membranes in high salinity conditions, with stability improving as DAS concentration increases.

Until now, we demonstrated the successful crosslinking of the PDADMAC/PSS multilayers under salt conditions up to 1.5 M. Next, we explored whether the stability of the membrane could be further extended to even higher salinities, up to 4 M NaCl. The results are presented in Fig. 9, illustrating the relative change in permeability during membrane filtration using NaCl solutions of varying ionic strengths and pure water as the feed. The non-crosslinked membrane showed a vast increase in permeability with increasing NaCl concentration. At 4 M NaCl, the permeability increased by 300 %, indicating significant swelling. The pure water permeability after exposure to solutions containing 3 M or 4 M NaCl of the non-crosslinked membrane

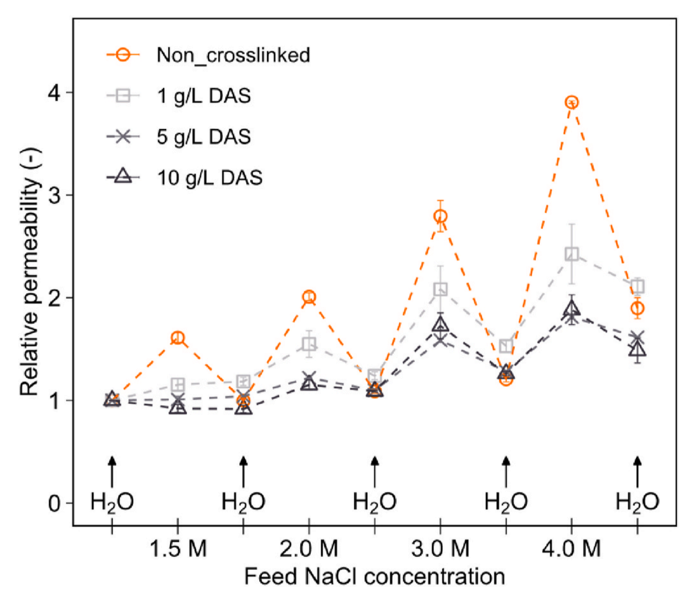

Fig. 9. Relative permeability (–) during filtration using pure water and NaCl solutions with ionic strength of 1.5 M, 2.0 M, 3.0 M, and 4.0 M, of $(PDADMAC/PSS)_{4.5}$-coated membranes without crosslinking as well as after crosslinking with DAS concentrations of 1 g L^{-1}, 5 g L^{-1} or 10 g L^{-1}. Error bars display the standard error (sample size n = 2). Relative permeability = (absolute permeability of a membrane sample measured at different conditions)/(pristine pure water permeability of the membrane sample).

also increased by 20 % and 110 %, respectively. This increase in pure water permeability suggests that the integrity of the membrane is (partly) compromised. The permeability change of the 1 g L^{-1} DAS crosslinked membrane showed a comparable trend, although the permeability increase in the presence of salt was less pronounced. In contrast, the 5 g L^{-1} and 10 g L^{-1} DAS crosslinked membranes demonstrated enhanced stability at higher ionic strengths, with a slight permeability increase of 15 %–20 % at 2 M NaCl. Even at 4 M NaCl, the permeability of these two membranes only increased by around 90 %, which was already the case for the non-crosslinked membrane at 1.5 M NaCl. These findings indicate that crosslinking can significantly reduce PEM swelling at high ionic strength. The stability of the crosslinked membranes may require further enhancement at NaCl concentrations above 3 M. However, they already show considerable potential for use in various applications that contain feed solutions of high ionic strengths.

4. Conclusions

This work demonstrates the effect of crosslinking on the stability of PDADMAC/PSS multilayers. By subjecting the multilayers to surfactants and monitoring in situ via reflectometry, it was shown that crosslinking with the photosensitive DAS significantly enhanced PEM stability. Building upon this, we developed stable PEM NF membranes. Crosslinking with DAS resulted in the formation of denser membranes with a lower MWCO, decreasing from 861 Da in non-crosslinked membranes to 354 Da in membranes crosslinked with a low DAS concentration (1 g L^{-1}). Furthermore, the incorporation of negatively charged DAS resulted in a shift of membrane charge from positive to highly negative, reflected by a high Na_2SO_4 retention of approximately 95 % and a low $CaCl_2$ retention of less than 7 % of the crosslinked membrane. As the concentration of DAS increases, more negative charges are incorporated, resulting in higher swelling of the PEM in water, which leads to membranes with enhanced water transport. The crosslinked membranes thus show promising separation properties with high selectivity and high permeability.

Stability studies indicate that crosslinking successfully enhanced membrane stability, maintaining consistent permeability up to 1.5 M NaCl. The non-crosslinked membrane showed relatively low stability, with a substantial increase in permeability when exposed to high salinity conditions, indicating high swelling and layer mobility. As anticipated, the crosslinked network greatly reduced layer swelling under high ionic strength conditions.

To summarize, for the first time, a crosslinked PDADMAC/PSS-based PEM NF membrane was developed, exhibiting full stability up to 1.5 M NaCl and even enhanced stability at higher ionic strengths, up to 4 M NaCl. An increase in DAS concentration from 1 g L^{-1} to 10 g L^{-1} further improved stability. The developed crosslinked membrane exhibits high separation efficiencies, highlighting crosslinking as a key tuning parameter for membrane density and charge properties. These findings reveal the strong potential of the crosslinked PEM NF membranes for various challenging process conditions where high salinity is a concern. In addition, we recommend future research to focus on aspects such as long-term membrane performance, and other harsh process conditions to further validate and broaden their applicability in practical applications.

CRediT authorship contribution statement

Xiao Zhang: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Antoine J.B. Kemperman:** Writing – review & editing, Supervision, Investigation, Conceptualization. **Henk Miedema:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Esra te Brinke:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Wiebe M. de Vos:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Wiebe M. de Vos reports financial support was provided by Wetsus. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.memsci.2025.124007.

Data availability

Data will be made available on request.

References

[1] E.N. Durmaz, S. Sahin, E. Virga, S. de Beer, L.C.P.M. de Smet, W.M. de Vos, Polyelectrolytes as building blocks for next-generation membranes with advanced functionalities, ACS Appl. Polym. Mater. 3 (2021) 4347–4374, https://doi.org/10.1021/acsapm.1c00654.

[2] X. Li, C. Liu, B. Van Der Bruggen, Polyelectrolytes self-Assembly: versatile membrane fabrication strategy, J Mater Chem A Mater 8 (2020) 20870–20896, https://doi.org/10.1039/d0ta07154d.

[3] G. Decher, Fuzzy nanoassemblies: toward layered polymeric multicomposites, Science 277 (1979) 1232–1237, 1997, http://www.jstor.org.ezproxy2.utwente.nl/stable/2892480.

[4] S. Lindhoud, W. Norde, M.A. Cohen Stuart, Reversibility and relaxation behavior of polyelectrolyte complex micelle formation, J. Phys. Chem. B 113 (2009) 5431–5439, https://doi.org/10.1021/jp809489f.

[5] J. Borges, J.F. Mano, Molecular interactions driving the layer-by-layer assembly of multilayers, Chem. Rev. 114 (2014) 8883–8942, https://doi.org/10.1021/cr400531v.

[6] J.B. Schlenoff, H. Ly, M. Li, Charge and mass balance in polyelectrolyte multilayers, J. Am. Chem. Soc. 120 (1998) 7626–7634, https://doi.org/10.1021/ja980350+.

[7] J. Kamp, S. Emonds, M. Seidenfaden, P. Papenheim, M. Kryschewski, J. Rubner, M. Wessling, Tuning the excess charge and inverting the salt rejection hierarchy of polyelectrolyte multilayer membranes, J. Membr. Sci. 639 (2021), https://doi.org/10.1016/j.memsci.2021.119636.

[8] Y. Wang, I. Zucker, C. Boo, M. Elimelech, Removal of emerging wastewater organic contaminants by polyelectrolyte multilayer nanofiltration membranes with tailored selectivity, ACS ES and T Engineering 1 (2021) 404–414, https://doi.org/10.1021/acsestengg.0c00160.

[9] E. te Brinke, D.M. Reurink, I. Achterhuis, J. de Grooth, W.M. de Vos, Asymmetric polyelectrolyte multilayer membranes with ultrathin separation layers for highly efficient micropollutant removal, Appl. Mater. Today 18 (2020), https://doi.org/10.1016/j.apmt.2019.100471.

[10] S.B. Rutten, V.L. Levering, L. Hernández Leal, J. de Grooth, H.D.W. Roesink, Retention of micropollutants by polyelectrolyte multilayer based hollow fiber nanofiltration membranes under fouled conditions, J. Water Proc. Eng. 53 (2023), https://doi.org/10.1016/j.jwpe.2023.103760.

[11] P. Ahmadiannamini, M.L. Bruening, V.V. Tarabara, Sacrificial polyelectrolyte multilayer coatings as an approach to membrane fouling control: disassembly and regeneration mechanisms, J. Membr. Sci. 491 (2015) 149–158, https://doi.org/10.1016/j.memsci.2015.04.041.
[12] O. Sanyal, Z. Liu, J. Yu, B.M. Meharg, J.S. Hong, W. Liao, I. Lee, Designing fouling-resistant clay-embedded polyelectrolyte multilayer membranes for wastewater effluent treatment, J. Membr. Sci. 512 (2016) 21–28, https://doi.org/10.1016/J.MEMSCI.2016.04.006.
[13] N. Wang, G. Zhang, S. Ji, Z. Qin, Z. Liu, The salt-, pH- and oxidant-responsive pervaporation behaviors of weak polyelectrolyte multilayer membranes, J. Membr. Sci. 354 (2010), https://doi.org/10.1016/j.memsci.2010.03.002.
[14] K. Glinel, C. Déjugnat, M. Prevot, B. Schöler, M. Schönhoff, R.v. Klitzing, Responsive polyelectrolyte multilayers, Colloids Surf. A Physicochem. Eng. Asp. 303 (2007) 3–13, https://doi.org/10.1016/j.colsurfa.2007.02.052.
[15] Y. Lu, Z. Qin, N. Wang, Q.F. An, H. Guo, Counterion exchanged hydrophobic polyelectrolyte multilayer membrane for organic solvent nanofiltration, J. Membr. Sci. 620 (2021), https://doi.org/10.1016/j.memsci.2020.118827.
[16] C. Wang, M.J. Park, D.H. Seo, S. Phuntsho, R.R. Gonzales, H. Matsuyama, E. Drioli, H.K. Shon, Inkjet printed polyelectrolyte multilayer membrane using a polyketone support for organic solvent nanofiltration, J. Membr. Sci. 642 (2022), https://doi.org/10.1016/j.memsci.2021.119943.
[17] K.L. Cho, H. Lomas, A.J. Hill, F. Caruso, S.E. Kentish, Spray assembled, cross-linked polyelectrolyte multilayer membranes for salt removal, Langmuir 30 (2014), https://doi.org/10.1021/la501855k.
[18] M.R. Moradi, A. Pihlajamäki, M. Hesampour, M. Figueira, M. Reig, J.L. Cortina, C. Valderrama, M. Mänttäri, Polyelectrolyte multilayers modification of nanofiltration membranes to improve selective separation of mono- and multivalent cations in seawater brine, J. Membr. Sci. 691 (2024), https://doi.org/10.1016/j.memsci.2023.122224.
[19] J. de Grooth, B. Haakmeester, C. Wever, J. Potreck, W.M. de Vos, K. Nijmeijer, Long term physical and chemical stability of polyelectrolyte multilayer membranes, J. Membr. Sci. 489 (2015) 153–159, https://doi.org/10.1016/j.memsci.2015.04.031.
[20] M.G. Elshof, W.M. de Vos, J. de Grooth, N.E. Benes, On the long-term pH stability of polyelectrolyte multilayer nanofiltration membranes, J. Membr. Sci. 615 (2020), https://doi.org/10.1016/j.memsci.2020.118532.
[21] J.M. Dickhout, J. Moreno, P.M. Biesheuvel, L. Boels, R.G.H. Lammertink, W.M. de Vos, Produced water treatment by membranes: a review from a colloidal perspective, J. Colloid Interface Sci. 487 (2017) 523–534, https://doi.org/10.1016/J.JCIS.2016.10.013.
[22] J. Chen, S. Xu, C.Y. Tang, B. Hu, B. Tokay, T. He, Stability of layer-by-layer nanofiltration membranes in highly saline streams, Desalination 555 (2023), https://doi.org/10.1016/j.desal.2023.116520.
[23] W. Cheng, C. Liu, T. Tong, R. Epsztein, M. Sun, R. Verduzco, J. Ma, M. Elimelech, Selective removal of divalent cations by polyelectrolyte multilayer nanofiltration membrane: role of polyelectrolyte charge, ion size, and ionic strength, J. Membr. Sci. 559 (2018) 98–106, https://doi.org/10.1016/j.memsci.2018.04.052.
[24] L. Han, Z. Mao, H. Wuliyasu, J. Wu, X. Gong, Y. Yang, C. Gao, Modulating the structure and properties of poly(sodium 4-styrenesulfonate)/poly(diallyldimethylammonium chloride) multilayers with concentrated salt solutions, Langmuir 28 (2012) 193–199, https://doi.org/10.1021/la2040533.
[25] M.A. Rahim, W.S. Choi, H.J. Lee, I.C. Jeon, Ionic surfactant-triggered renewal of the structures and properties of polyelectrolyte multilayer films, Langmuir 26 (2010) 4680–4686, https://doi.org/10.1021/la9035634.
[26] E. Virga, J. de Grooth, K. Žvab, W.M. de Vos, Stable polyelectrolyte multilayer-based hollow fiber nanofiltration membranes for produced water treatment, ACS Appl. Polym. Mater. 1 (2019) 2230–2239, https://doi.org/10.1021/acsapm.9b00503.
[27] P.H.H. Duong, J. Zuo, T.S. Chung, Highly crosslinked layer-by-layer polyelectrolyte FO membranes: understanding effects of salt concentration and deposition time on FO performance, J. Membr. Sci. 427 (2013) 411–421, https://doi.org/10.1016/j.memsci.2012.10.014.
[28] K.L. Cho, A.J. Hill, F. Caruso, S.E. Kentish, Chlorine resistant glutaraldehyde crosslinked polyelectrolyte multilayer membranes for desalination, Adv. Mater. 27 (2015) 2791–2796, https://doi.org/10.1002/adma.201405783.
[29] S. Xu, R. He, C. Dong, N. Sun, S. Zhao, H. He, H. Yu, Y.B. Zhang, T. He, Acid stable layer-by-layer nanofiltration membranes for phosphoric acid purification, J. Membr. Sci. 644 (2022), https://doi.org/10.1016/j.memsci.2021.120090.
[30] W. Tong, C. Gao, H. Möhwald, Manipulating the properties of polyelectrolyte microcapsules by glutaraldehyde cross-linking, Chem. Mater. 17 (2005) 4610–4616, https://doi.org/10.1021/cm0507516.
[31] M. Schock, S. Bräse, Reactive & efficient: organic azides as cross-linkers in material sciences, Molecules 25 (2020), https://doi.org/10.3390/molecules25041009.
[32] H. Liu, H. Ruan, Y. Zhao, J. Pan, A. Sotto, C. Gao, B. van der Bruggen, J. Shen, A facile avenue to modify polyelectrolyte multilayers on anion exchange membranes to enhance monovalent selectivity and durability simultaneously, J. Membr. Sci. 543 (2017) 310–318, https://doi.org/10.1016/j.memsci.2017.08.072.
[33] M.A. Restrepo, S. Emonds, A. Zhao, F. Karakas, J. Kamp, H. Roth, M. Wessling, Self-supporting biocatalytic polyelectrolyte complex hollow fiber membranes via salt-dilution induced phase separation, J. Membr. Sci. 689 (2024), https://doi.org/10.1016/j.memsci.2023.122157.
[34] S.D. Tuong, H. Lee, H. Kim, Chemical fixation of polyelectrolyte multilayers on polymer substrates, Macromol. Res. 16 (2008), https://doi.org/10.1007/BF03218531.
[35] N.G. Hoogeveen, M.A. Cohen Stuart, G.J. Fleer, M.R. Böhmer, Formation and stability of multilayers of polyelectrolytes, Langmuir 12 (1996) 3675–3681, https://doi.org/10.1021/la951574y.
[36] J.C. Dijt, M.A.C. Stuart, J.E. Hofman, G.J. Fleer, Kinetics of polymer adsorption in stagnation point flow, Colloids Surf. 51 (1990), https://doi.org/10.1016/0166-6622(90)80138-T.
[37] B. Naskar, A. Dey, S.P. Moulik, Counter-ion effect on micellization of ionic surfactants: a comprehensive understanding with two representatives, sodium dodecyl sulfate (SDS) and dodecyltrimethylammonium bromide (DTAB), J. Surfactants Deterg. 16 (2013) 785–794, https://doi.org/10.1007/s11743-013-1449-1.
[38] J.C. Dijt, M.A.C. Stuart, G.J. Fleer, Reflectometry as a tool for adsorption studies, Adv. Colloid Interface Sci. 50 (1994), https://doi.org/10.1016/0001-8686(94)80026-X.
[39] W. Ogieglo, H. Wormeester, K.J. Eichhorn, M. Wessling, N.E. Benes, In situ ellipsometry studies on swelling of thin polymer films: a review, Prog. Polym. Sci. 42 (2015) 42–78, https://doi.org/10.1016/j.progpolymsci.2014.09.004.
[40] G.R. Guillen, Y. Pan, M. Li, E.M.V. Hoek, Preparation and characterization of membranes formed by nonsolvent induced phase separation: a review, Ind. Eng. Chem. Res. 50 (2011) 3798–3817, https://doi.org/10.1021/ie101928r.
[41] N. Schonbeck, K. Kvale, T. Demarcy, J. Giermanska, J.-P. Chapel, J.-F. Berret, Surfactant-Triggered disassembly of electrostatic complexes probed at optical and quartz crystal microbalance length scales, Langmuir 30 (2014) 5620–5627, https://doi.org/10.1021/la500948h.
[42] J.J.I. Ramos, I. Llarena, L. Fernández, S.E. Moya, E. Donath, Controlled stripping of polyelectrolyte multilayers by quaternary ammonium surfactants, Macromol. Rapid Commun. 30 (2009), https://doi.org/10.1002/marc.200900280.
[43] M.D. Miller, M.L. Bruening, Correlation of the swelling and permeability of polyelectrolyte multilayer films, Chem. Mater. 17 (2005) 5375–5381, https://doi.org/10.1021/cm0512225.
[44] T. Takamatsu, M. Hashiyama, A. Eisenberg, Sorption phenomena in nafion membranes, J. Appl. Polym. Sci. 24 (1979), https://doi.org/10.1002/app.1979.070241101.
[45] M. McCormick, R.N. Smith, R. Graf, C.J. Barrett, L. Reven, H.W. Spiess, NMR studies of the effect of adsorbed water on polyelectrolyte multilayer films in the solid state, Macromolecules 36 (2003), https://doi.org/10.1021/ma034147x.
[46] J. de Grooth, R. Oborný, J. Potreck, K. Nijmeijer, W.M. de Vos, The role of ionic strength and odd-even effects on the properties of polyelectrolyte multilayer nanofiltration membranes, J. Membr. Sci. 475 (2015) 311–319, https://doi.org/10.1016/j.memsci.2014.10.044.
[47] R.A. Ghostine, M.Z. Markarian, J.B. Schlenoff, Asymmetric growth in polyelectrolyte multilayers, J. Am. Chem. Soc. 135 (2013) 7636–7646, https://doi.org/10.1021/ja401318m.
[48] J.M.M. Peeters, J.P. Boom, M.H. V Mulder, H. Strathmann, Retention measurements of nanofiltration membranes with electrolyte solutions, J. Membr. Sci. 145 (1998) 199–209, https://doi.org/10.1016/S0376-7388(98)00079-9.
[49] A. Yaroshchuk, M.L. Bruening, E. Zholkovskiy, Modelling nanofiltration of electrolyte solutions, Adv. Colloid Interface Sci. 268 (2019) 39–63, https://doi.org/10.1016/j.cis.2019.03.004.
[50] W.R. Bowen, H. Mukhtar, Characterisation and Prediction of Separation Performance of Nanofiltration Membranes, 1996.
[51] C. Zhong, Y. Deng, W. Hu, J. Qiao, L. Zhang, J. Zhang, A review of electrolyte materials and compositions for electrochemical supercapacitors, Chem. Soc. Rev. 44 (2015) 7484–7539, https://doi.org/10.1039/c5cs00303b.
[52] K. Boussu, C. Vandecasteele, B. Van der Bruggen, Relation between membrane characteristics and performance in nanofiltration, J. Membr. Sci. 310 (2008) 51–65, https://doi.org/10.1016/j.memsci.2007.10.030.
[53] J.A. Jaber, J.B. Schlenoff, Counterions and water in polyelectrolyte multilayers: a tale of two polycations, Langmuir 23 (2007) 896–901, https://doi.org/10.1021/la061839g.
[54] S.T. Dubas, J.B. Schlenoff, Swelling and smoothing of polyelectrolyte multilayers by salt, Langmuir 17 (2001) 7725–7727, https://doi.org/10.1021/la0112099.
[55] S. Lindhoud, W. Norde, M.A. Cohen Stuart, Reversibility and relaxation behavior of polyelectrolyte complex micelle formation, J. Phys. Chem. B 113 (2009) 5431–5439, https://doi.org/10.1021/jp809489f.